Second Midterm Exam

- The third midterm exam will be given on Nov 20 in class.
- The exam is based on Lectures 1–16 (including today).

Recall : definition of determinant

What is det A?

Suppose A is an $n \times n$ matrix.

- The determinant of A is a number which can be calculated by cofactor expansion across a row or along a column.
- The main property of det A: A is invertible if and only if $\det A \neq 0$.

Determinant of a 2×2 matrix

If $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ then

$$\det A = a_{11}a_{22} - a_{12}a_{21}$$

Sometimes we write this as $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$

Determinant of a 3×3 matrix - row expansions

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \qquad \rightsquigarrow \qquad \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

To compute the determinant we can use cofactor expansions across a row or along a column.

- **Cofactor expansion across the first row :**

$$\det A = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

- **Cofactor expansion along the second column :**

$$\det A = -a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{22} \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} - a_{32} \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix}$$

The above method generalizes to $n \times n$ matrices.

Another method of computing det A

Let A be an $n \times n$ matrix. Recall that we have three kinds of elementary operations :

1. Interchange : $R_i \leftrightsquigarrow R_j$.
2. Replacement : $R_i \to R_i + cR_j$.
3. Scaling : $R_i \to cR_i$.

- To calculate an E.F. of A, we only need Interchange and Replacement. The Scaling operation is only required when we want to obtain the R.E.F.
- Replacement does not change the determinant. Interchange changes the sign of the determinant.
- Suppose we row reduce A, using only interchanges and replacements, to obtain an echelon form :

$$A \xrightarrow{\text{Row Reduction}} B \qquad \text{E.F.}$$

Then

$$\det A = (-1)^r \det B$$

where r is the number of interchanges done throughout the reduction process.

Example

Calculate det A where $A = \begin{bmatrix} 2 & -1 & 0 & 1 & 2 \\ 2 & -1 & -1 & 3 & 2 \\ 0 & 1 & 3 & 0 & 1 \\ 0 & 2 & 6 & 3 & 3 \\ 2 & 0 & 3 & 1 & 7 \end{bmatrix}$

We row reduce to obtain an E.F. :

$$\begin{vmatrix} 2 & -1 & 0 & 1 & 2 \\ 2 & -1 & -1 & 3 & 2 \\ 0 & 1 & 3 & 0 & 1 \\ 0 & 2 & 6 & 3 & 3 \\ 2 & 0 & 3 & 1 & 7 \end{vmatrix} \xrightarrow[R_5 \to R_5 - R_1]{R_2 \to R_2 - R_1} \begin{bmatrix} 2 & -1 & 0 & 1 & 2 \\ 0 & 0 & -1 & 2 & 0 \\ 0 & 1 & 3 & 0 & 1 \\ 0 & 2 & 6 & 3 & 3 \\ 0 & 1 & 3 & 0 & 5 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 2 & -1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 0 & 1 \\ 0 & 0 & -1 & 2 & 0 \\ 0 & 2 & 6 & 3 & 3 \\ 0 & 1 & 3 & 0 & 5 \end{bmatrix}$$

$$\xrightarrow[R_5 \to R_5 - R_2]{R_4 \to R_4 - 2R_2} \begin{bmatrix} 2 & -1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 0 & 1 \\ 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix} \quad \text{E.F.}$$

The last matrix is upper triangular and its determinant is $(2)(1)(-1)(3)(4) = -24$.

Since there was only one interchange in the above reduction process, we have

$$\det A = (-1)^1 \cdot (-24) = 24$$

Example

Calculate $\det A$ where $A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 1 & 2 & 5 & 1 \\ 0 & 3 & -1 & 3 \\ 2 & 1 & 0 & 1 \end{bmatrix}$.

Solution.

$$\begin{vmatrix} 1 & 3 & 1 & 2 \\ 1 & 2 & 5 & 1 \\ 0 & 3 & -1 & 3 \\ 2 & 1 & 0 & 1 \end{vmatrix} \xrightarrow[R_4 \to R_4 - 2R_1]{R_2 \to R_2 - R_1} \begin{vmatrix} 1 & 3 & 1 & 2 \\ 0 & -1 & 4 & -1 \\ 0 & 3 & -1 & 3 \\ 0 & -5 & -2 & -3 \end{vmatrix}$$

$$\xrightarrow[R_4 \to R_4 - 5R_2]{R_3 \to R_3 + 3R_2} \begin{vmatrix} 1 & 3 & 1 & 2 \\ 0 & -1 & 4 & -1 \\ 0 & 0 & 11 & 0 \\ 0 & 0 & -22 & 2 \end{vmatrix} \xrightarrow{R_4 \to R_4 + 2R_3} \begin{vmatrix} 1 & 3 & 1 & 2 \\ 0 & -1 & 4 & -1 \\ 0 & 0 & 11 & 0 \\ 0 & 0 & 0 & 2 \end{vmatrix} = B \quad \text{E.F.}$$

We didn't use any interchange. Therefore $\det A = \det B = -22$.

Let A be an $n \times n$ matrix.

If A has a zero row or a zero column, then $\det A = 0$.

Example

$$\begin{vmatrix} 1 & 0 & 1 & 4 \\ -2 & 0 & 1 & -5 \\ \frac{1}{2} & 0 & 8 & -3 \\ -2 & 0 & -4 & 5 \end{vmatrix} = 0 \quad \text{and} \quad \begin{vmatrix} -3 & 4 & 1 & 12 \\ 2 & 0 & 1 & -5 \\ 0 & 0 & 0 & 0 \\ -2 & 6 & -4 & 5 \end{vmatrix} = 0$$

- If two rows of A are identical, then $\det A = 0$.
- If two columns of A are identical, then $\det A = 0$.

Example

$$\begin{vmatrix} -3 & 4 & 1 & 12 \\ 2 & 0 & 1 & -5 \\ -3 & 4 & 1 & 12 \\ -2 & 6 & -4 & 5 \end{vmatrix} = 0$$

Theorem

For a square matrix A we have $\det A^T = \det A$.

Theorem

If A and B are $n \times n$ matrices, then

$$\det(AB) = (\det A)(\det B)$$

Theorem

If A is invertible, then $\det(A^{-1}) = \dfrac{1}{\det A}$

Reason: $AA^{-1} = \mathrm{I} \quad \Longrightarrow \quad \Big(\det A\Big)\Big(\det(A^{-1})\Big) = \det \mathrm{I} = 1.$

Roots of equations

- Consider the equation $x^2 - 1 = 0$. Solve it!

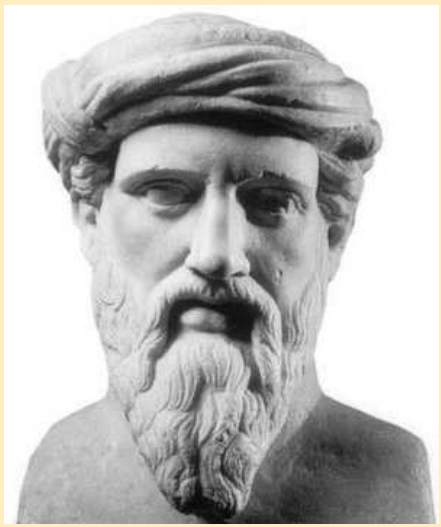
  $x^2 = 1 \quad \Longrightarrow \quad x = \pm 1.$
- How about $4x^2 - 1 = 0$?

  $x^2 = \frac{1}{4} \quad \Longrightarrow \quad x = \pm \frac{1}{2}$
- How about $x^2 - 2 = 0$?

  $x^2 = 2 \quad \Longrightarrow \quad x = \pm \sqrt{2}$

History of $\sqrt{2}$

- $\sqrt{2} = 1.41421356237309504880168887 \cdots$
- $\sqrt{2} \approx \frac{7}{5}$, $\sqrt{2} \approx \frac{17}{12}$, $\sqrt{2} \approx \frac{577}{408}$
- Pythagoras (circa 570 B.C.– circa 495 B.C.) was among the first people who discovered that $\sqrt{2}$ cannot be expressed as a ratio of two integers.
- The repercussions of this discovery might have led to the demise of Pythagoras' school of philosophical thought.



Complex numbers - History

- How about $x^2 + 1 = 0$?

  We can write it as $x^2 = -1$ and obviously, there is no *real* number x whose square is negative.

- BUT...we can introduce a new "number" i such that $i^2 + 1 = 0$.

- This novel idea was first introduced by the Italian polymath Gerolamo Cardano (1501 – 1576)

- He discovered new *imaginary* numbers.



The number "i"

- The equation $x^2 = -1$ has two roots, which are not real numbers. They are represented by "i" and "$-i$".

- Therefore i and $-i$ are two imaginary numbers such that

$$i^2 = -1 \quad \text{and} \quad (-i)^2 = (-1)^2(i)^2 = 1 \times (-1) = -1.$$

Complex numbers

A complex number is a number written in the form

$$a + bi$$

where :

- a and b are ordinary "real" numbers,
- i is the imaginary number we just discussed, i.e., $i^2 = -1$.

Let $z = a + bi$. Then :

- a is called the real part of z and is represented by $\operatorname{Re} z$
- b is called the imaginary part of z and is represented by $\operatorname{Im} z$.

Example

- $z = 3 - 5i$ is a complex number. We have $\operatorname{Re} z = 3$ and $\operatorname{Im} z = -5$.
- $z = -1 + i$ is a complex number. We have $\operatorname{Re} z = -1$ and $\operatorname{Im} z = 1$.
- $-i$ a complex number. We have $\operatorname{Re} z = 0$ and $\operatorname{Im} z = -1$.

Real vs. Complex Numbers

- Note that if x is a *real* number then x is also a complex number ($\operatorname{Re} x = x$ and $\operatorname{Im} x = 0$).
- A complex number of the form $0 + bi$ is sometimes called an imaginary number.

  Example. $-3i$, i, $\frac{1}{3}i$ and $-\sqrt{2}i$ are imaginary numbers.

- The set of complex numbers is denoted by $\mathbb{C}$. (Recall that we denoted the set of real numbers by $\mathbb{R}$.)
- Therefore $\mathbb{R}$ is a subset of $\mathbb{C}$!

Addition of complex numbers

How do we add complex numbers?

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

Example

- Suppose we want to add $3 - i$ and $1 + 2i$.

$$(3 - i) + (1 + 2i) = (3 + 1) + (-1 + 2)i = 4 + i$$

- Suppose we want to add $-1 + i$ and $2 + 3i$.

$$(-1 + i) + (2 + 3i) = (-1 + 2) + (1 + 3)i = 1 + 4i$$

- Suppose we want to add $i - 2$ and $1 - 2i$.

We first note that $i - 2 = -2 + i$. Then :

$$(-2 + i) + (1 - 2i) = (-2 + 1) + (1 - 2)i = -1 - i$$

Subtraction of complex numbers

$$(a + bi) - (c + di) = (a - c) + (b - d)i$$

Example

$(3 - 5i) - (2 - 4i) = (3 - 2) + (-5 - (-4))i = 1 - i$

Multiplication of complex numbers

How do we multiply complex numbers?

$$(a+bi)(c+di) = ac + adi + bic + bidi = ac + adi + bci + bdi^2 = (ac - bd) + (ad + bc)i$$

Example

$$(2-3i)(-1+2i) = 2(-1) + 2(2)i + (-3)(-1)i + (-3)(2)i^2 = -2 + 4i + 3i + 6 = 4 + 7i$$

$$(i-1)(3-i) = 3i + i(-i) + (-1)(3) + (-1)(-i) = 3i - i^2 - 3 + i = -2 + 4i$$

$$3(2-3i) = 6 - 9i.$$

$$i^4 = i \cdot i \cdot i \cdot i = (-1) \cdot i \cdot i = (-1)(-1) = 1.$$

Example

- Solve the equation $x^2 = -4$.

$x^2 + 4 = 0 \qquad \Longrightarrow \qquad (x + 2i)(x - 2i) = 0 \qquad \Longrightarrow \qquad x = \pm 2i$

In general, if $r > 0$, then $x^2 + r = (x + \sqrt{r}\, i)(x - \sqrt{r}\, i)$.

Complex conjugation

For a complex number $z = a + bi$, the complex conjugate of z is defined as

$$\overline{z} = a - bi$$

Example

$\overline{3 - 2i} = 3 + 2i$

$\overline{\frac{1}{2} + \sqrt{3}i} = \frac{1}{2} - \sqrt{3}i$

$\overline{-2} = -2$

Absolute value of a real number

Recall that for a real number x, the absolute value $|x|$ is defined by:

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

For example:

$|3| = 3 \qquad |\frac{3}{5}| = \frac{3}{5} \qquad |2.5| = 2.5$

while:

$|-3| = 3 \qquad |-.5| = .5 \qquad |-\frac{1}{3}| = \frac{1}{3}.$

Absolute value of a complex number

Consider a complex number $z = a + bi$. The absolute value (or modulus or norm) of z is defined by

$$|z| = \sqrt{a^2 + b^2}$$

Note that $z\overline{z} = (a + bi)(a - bi) = a^2 + b^2$. Therefore we have

$$|z| = \sqrt{z\overline{z}}$$

- The absolute value of a complex number is always a nonnegative real number.

Example

$|3 - 4i| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$

$|-1 - 2i| = \sqrt{(-1)^2 + (-2)^2} = \sqrt{5}$

Note that if a is a real number then:

$$|ai| = \sqrt{a^2} = |a|$$

For instance $|-5i| = \sqrt{(-5)^2} = \sqrt{25} = 5.$

Basic properties of complex numbers

Let z, z_1, and z_2 be complex numbers. Then :

1. $\overline{z} = z$ if and only if z is a real number, i.e., $\operatorname{Im} z = 0$.
2. $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$
3. $\overline{z_1 z_2} = \overline{z_1}\,\overline{z_2}$
4. $z\overline{z} = |z|^2 \geqslant 0$.
5. $|z_1 z_2| = |z_1||z_2|$
6. Triangle Inequality : $|z_1 + z_2| \leqslant |z_1| + |z_2|$

Every polynomial equation $a_0 + a_1 x + \cdots + a_n x^n = 0$ with $a_0, \ldots, a_n \in \mathbb{C}$ and $n > 0$ has a solution/root in $\mathbb{C}$.

Division of complex numbers

Example

Simplify $\dfrac{1+3i}{-1+5i}$.

Note that

$$\frac{1+3i}{-1+5i} = \frac{(1+3i)(-1-5i)}{(-1+5i)(-1-5i)} = \frac{-1-5i-3i-15i^2}{1+5i-5i-25i^2} = \frac{14-8i}{26}$$

$$= \frac{1}{26}(14-8i) = \frac{14}{26} - \frac{8}{26}i$$

Division of complex numbers

In general, if $z = a + bi$ and $w = c + di$ are complex numbers and $w \neq 0$, then $\frac{z}{w}$ is obtained as follows :

$$\frac{z}{w} = \frac{z\overline{w}}{w\overline{w}} = \frac{1}{|w|^2}(z\overline{w})$$

Example

Simplify $\dfrac{3-i}{-1+i}$.

$$\frac{3-i}{-1+i} = \frac{(3-i)(-1-i)}{(-1+i)(-1-i)} = \frac{-3-3i-(-i)+(-i)^2}{(-1)^2+(-1)(-i)+i(-1)+i(-i)} = \frac{-4-2i}{2} = -2-i$$

In particular, one can see that if $z = a + bi$ is a complex number then

$$\frac{1}{z} = \frac{a}{a^2+b^2} - \frac{b}{a^2+b^2}i$$

Example

Simplify $\dfrac{1}{-2+i}$.

Solution. $a = -2, b = 1 \Rightarrow \dfrac{1}{-2+i} = \dfrac{-2}{((-2)^2+1)} - \dfrac{1}{(-2)^2+1}i = -\dfrac{2}{5} - \dfrac{1}{5}i.$

We can check that:

$$(-2+i)(-\frac{2}{5} - \frac{1}{5}i) = 1$$

Example

Solve the equation $(1+2i)z + 2 = \dfrac{2}{1+i}$.

Solution.

$$(1+2i)z = \frac{2}{1+i} - 2 = \frac{2-2(1+i)}{1+i} = \frac{2-2-2i}{1+i} = \frac{-2i}{1+i}$$

$$z = \frac{\frac{-2i}{1+i}}{1+2i} = \frac{-2i}{(1+i)(1+2i)} = \frac{-2i}{1+3i+2i^2} = \frac{-2i}{-1+3i} = \frac{-2i(-1-3i)}{(-1+3i)(-1-3i)}$$

$$= \frac{2i+6i^2}{(1-3i+3i-9i^2)} = \frac{-6+2i}{10} = -\frac{3}{5} + \frac{1}{5}i$$

$\Rightarrow z = -\frac{3}{5} + \frac{1}{5}i.$

- We can now do everything we have done with vectors and matrices using complex numbers.

Example

$$\begin{bmatrix} 1 & 1-i \\ -i & 2 \end{bmatrix}\begin{bmatrix} -1 & i \\ 0 & i+1 \end{bmatrix} = \begin{bmatrix} 1(-1)+(1-i)(0) & 1(i)+(1-i)(i+1) \\ (-i)(-1)+2(0) & (-i)i+2(i+1) \end{bmatrix} = \begin{bmatrix} -1 & 2+i \\ i & 2+2i \end{bmatrix}$$

Example

Find the inverse of

$$A = \begin{bmatrix} 1 & i \\ -1 & i \end{bmatrix}$$

Solution. $\det A = 1(i) - i(-1) = 2i \neq 0$. Therefore:

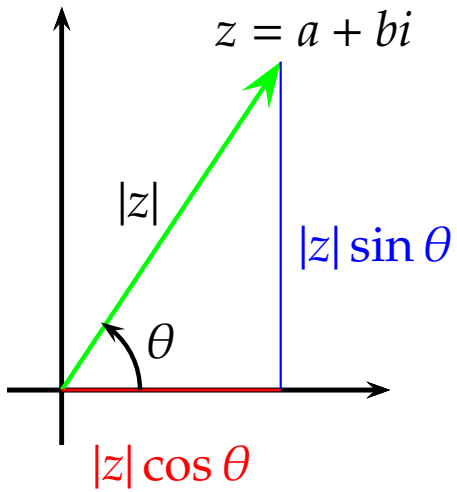
$$A^{-1} = \frac{1}{2i}\begin{bmatrix} i & -i \\ 1 & 1 \end{bmatrix} = (-\frac{1}{2}i)\begin{bmatrix} i & -i \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2}i & -\frac{1}{2}i \end{bmatrix}$$


DO i REALLY EXIST?
√-1
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Geometric interpretation of complex numbers

How can we "see" complex numbers?

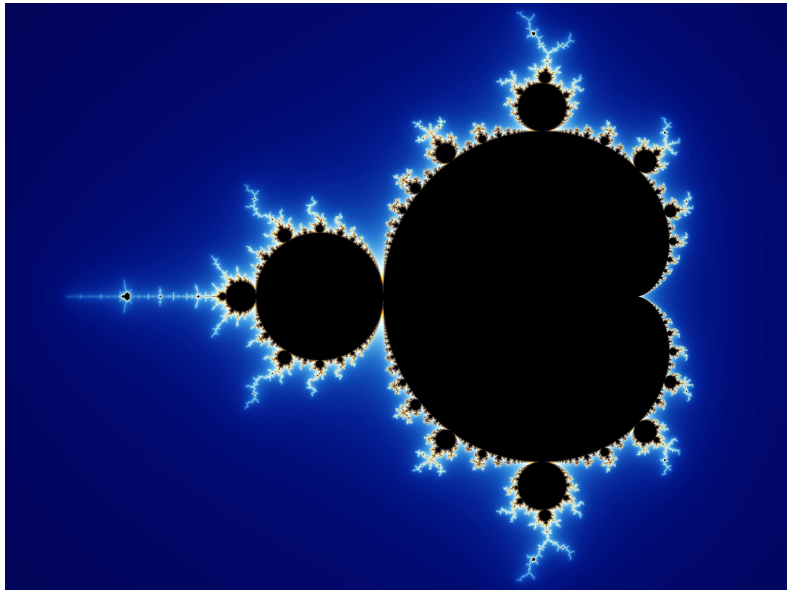
- A complex number $z = a + bi$ is represented by the vector $\begin{bmatrix} a \\ b \end{bmatrix}$ in the xy-plane.



- One can see that $z = a + bi = |z| \cos\theta + (|z| \sin\theta)i = |z|(\cos\theta + i \sin\theta)$

$$\begin{cases} a = |z| \cos\theta \\ b = |z| \sin\theta \end{cases}$$

The Mandelbrot Set



- A simple formula as simple as $f(z) = z^2 + c$, plus a bit of mathematicians' ingenuity, creates mysterious, substantially complex, mesmerizing images.

http://www.youtube.com/watch?v=9G6uO7ZHtK8& feature=related